

Md Rashedur Rahman

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ACADEMIC PROFILE

Ph.D. Candidate in Information Technology and M.S. graduate in Applied Information Technology with academic, research, and professional experience in information systems, software development, cybersecurity, networking, and operating systems. Graduate Teaching Assistant experienced in mentoring students, evaluating technical work, and providing constructive feedback. Brings nearly five years of professional experience developing mobile applications, web applications, APIs, reporting tools, and IT systems in public-health, financial-services, and banking environments. Committed to hands-on, student-centered IT education in networking, cybersecurity, databases, cloud and virtual environments, operating systems, and web/mobile development.

EDUCATION

George Mason University <i>Ph.D. Candidate in Information Technology (Expected Spring/Summer 2027)</i>	Fairfax, VA, USA Aug. 2022 – Present
George Mason University <i>M.S. in Applied Information Technology (Cyber Security Concentration)</i>	Fairfax, VA, USA Aug. 2022 – Dec. 2024
Islamic University of Technology <i>B.Sc. in Computer Science</i>	Gazipur, Bangladesh Dec. 2011 – Nov. 2015

TEACHING AND STUDENT MENTORING EXPERIENCE

Graduate Teaching Assistant <i>Department of Cyber Security Engineering, George Mason University</i>	Aug. 2023 – Present Fairfax, VA, USA
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- Supported undergraduate and graduate instruction in Computer Networking, Operating Systems, Secure RF Communications, Industrial Control Systems Security, and Critical Infrastructure Protection.
- Evaluated assignments, laboratory work, technical reports, and presentations; provided feedback to strengthen students' technical understanding, troubleshooting, and problem-solving skills.
- Mentored students through office hours and individual consultations on networking, Linux and operating-system concepts, wireless systems, cybersecurity, assignments, and technical troubleshooting.
- Supported accelerated summer offerings of CYSE 230: Computer Networking and CYSE 425: Secure RF Communications through grading, student communication, and instructional coordination.
- Collaborated with course instructors to address student concerns, support course operations, and maintain an inclusive, student-centered learning environment.

TEACHING AREAS

Information Technology and Infrastructure: Computer Networking; Operating Systems; Linux Systems; Cybersecurity Fundamentals; Cloud Computing and Virtualization; Wireless and Mobile Systems.

Information Systems and Applications: Information Systems Analysis and Design; Database Management Systems; Web and Mobile Application Development; Programming in Java and Python; Software Engineering; Internet of Things.

PROFESSIONAL EXPERIENCE

Programmer <i>icddr,b</i>	Jan. 2020 – Jan. 2021 Dhaka, Bangladesh
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- Developed Android-based data-collection applications for rural health workers supporting public-health studies related to emerging infectious diseases, including COVID-19 and Nipah-related projects.
- Built reporting tools that enabled researchers to retrieve, monitor, and analyze field-collected data efficiently.
- Provided technical support for field-data collection applications and contributed to improvements in research information systems.

Senior Officer, IT <i>Impress Capital Limited</i>	Feb. 2018 – Jan. 2020 Dhaka, Bangladesh
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- Developed Android and iOS applications that enabled investors to access portfolio information.
- Built APIs, web applications, and internal automation tools to improve operational workflows.
- Supported the development and improvement of internal IT systems and secure server operations.

Software Engineer

i-clique Solutions Limited

Sep. 2016 – Sep. 2017

Dhaka, Bangladesh

- Contributed to the development and customization of CardPro credit-card management software for banking clients in Bangladesh, Pakistan, and Mongolia.
- Supported technical onboarding and training for new team members working with the existing software system.

Senior Data Management Assistant (Programmer)

icddr, b

Dec. 2015 – Aug. 2016

Dhaka, Bangladesh

- Developed reporting tools that helped research teams monitor field-data collection for public-health-related projects and contributed to web-application development.

RESEARCH EXPERIENCE

Graduate Research Assistant

George Mason University

Aug. 2022 – Present

Fairfax, VA, USA

- Conduct interdisciplinary research at the intersection of cybersecurity, wireless communication, and IoT, focusing on security and resilience in 5G/6G, Wi-Fi, and shared-spectrum systems.
- Design and evaluate attack models, anomaly-detection methods, and resilience frameworks through simulation and wireless testbed experiments.
- Develop research tools and experimental workflows using Python, ns-3, SimPy, MATLAB, OpenAirInterface, and software-defined radio platforms.

Graduate Assistant

Towson University

Jan. 2021 – Aug. 2022

Towson, MD, USA

- Investigated privacy and informed-consent issues in Android permission models and disclosure practices in social-media privacy policies.
- Developed simulation environments to analyze security vulnerabilities in spectrum coexistence among heterogeneous wireless technologies.

Independent Researcher — Healthcare Wireless Sensor Networks

Independent Research Project

Jan. 2018 – Jun. 2021

Gazipur, Bangladesh

- Designed and evaluated application-specific routing mechanisms for healthcare wireless sensor networks, emphasizing reliable data delivery, energy efficiency, network lifetime, and throughput.
- Developed the ASRPH framework for healthcare sensing applications and compared its performance with established wireless-sensor-network routing approaches.
- Produced two peer-reviewed publications, including an IEEE conference paper and a *Computer Networks* journal article.

TECHNICAL SKILLS

Programming, Web, and Mobile Development: Python, Java, C++, C#, Swift, SQL, HTML/CSS, Android SDK, iOS/Swift Development, RESTful APIs

Information Systems and Databases: MySQL, PostgreSQL, Oracle, Microsoft SQL Server, Object-Oriented Programming, Data Structures, Algorithms

Systems, Networking, and Security: Linux (CLI), Windows, TCP/IP, DNS, VPN, Wi-Fi, Wireshark, Bash, PowerShell

Cloud, Research, and Development Tools: Git/GitHub, ns-3, OMNeT++, SimPy, MATLAB, OpenAirInterface

PEER-REVIEWED PUBLICATIONS

M. R. Rahman and P. Costa, “Ghost Rider: Anomaly-Aware Hybrid Reservation and Self-Deferral Framework for 5G/B5G Unlicensed in the 5 GHz Spectrum Band,” *Proceedings of the 2026 IEEE International Conference on Communications (ICC)*, 2026. [Preprint]

M. R. Rahman and P. Costa, “Z-Force: A Z-Score-Powered Anomaly Detection Framework for Fair 5G NR-U & Wi-Fi Coexistence in the 5 GHz Band,” *Proceedings of the 2025 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2025. [DOI]

M. R. Rahman and M. Hossain, “Channel Access Deterrence Attack: An Attack against Spectrum Coexistence between NR-U and Wi-Fi in the 5 GHz Band,” *Proceedings of the IEEE International Conference on Computer Communications (INFOCOM)*, 2025, pp. 1–10. [DOI]

M. R. Rahman and M. Hossain, “RanCAD: Random Channel Access Deterrence Attack against Spectrum Coexistence between NR-U and Wi-Fi on the 5 GHz Unlicensed Band,” *Proceedings of the 2024 IEEE International Conference on Communications (ICC)*, 2024. [DOI]

M. R. Rahman, M. Hossain, and J. Xie, “PACMAN Attack: A Mobility-Powered Attack in Private 5G-Enabled Industrial Automation System,” *Proceedings of the 2023 IEEE International Conference on Communications (ICC)*, 2023, pp. 4379–4384. [DOI]

E. Miller, **M. R. Rahman**, M. Hossain, and A. I. Ali-Gombe, “I Don’t Know Why You Need My Data: A Case Study of Popular Social Media Privacy Policies,” *Proceedings of the 12th ACM Conference on Data and Application Security and Privacy (CODASPY)*, 2022, pp. 340–342. [DOI]

M. R. Rahman, E. Miller, M. Hossain, and A. I. Ali-Gombe, “Intent-Aware Permission Architecture: A Model for Rethinking Informed Consent for Android Apps,” *Proceedings of the 8th International Conference on Information Systems Security and Privacy (ICISSP)*, 2022, pp. 154–164. [DOI]

M. R. Rahman, M. M. Islam, A. I. Pritom, and Y. Alsaawy, “ASRPH: Application Specific Routing Protocol for Health Care,” *Computer Networks*, vol. 197, Art. 108273, 2021. [DOI]

A. I. Pritom, M. Z. Chowdhury, J. Protim, S. Roy, **M. R. Rahman**, and S. M. Promi, “Combining Movement Model with Finger-Stroke Level Model towards Designing a Security Enhancing Mobile Friendly CAPTCHA,” *Proceedings of the 2020 9th International Conference on Software and Computer Applications*, 2020, pp. 351–356. [DOI]

M. T. Hosain, A. Al Arif, A. I. Pritom, **M. R. Rahman**, and M. Z. Islam, “Development of a Tangent Based Robust Speech Feature Extraction Model,” *Cyber Security and Computer Science (ICONCS 2020)*, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol. 325, Springer, 2020, pp. 414–425. [DOI]

M. R. Rahman, M. M. Islam, E. A. Shahaz, and Y. Alsaawy, “Application Specific Energy Aware and Reliable Routing Protocol for Wireless Sensor Network,” *Proceedings of the 2019 7th International Conference on Smart Computing & Communications (ICSCC)*, 2019, pp. 1–5. [DOI]

PROFESSIONAL SERVICE AND RECOGNITION

Peer Review Service: Peer Reviewer, *IEEE Transactions on Mobile Computing* (2025); Review Committee Member, International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2026); Review Committee Member, International Conference on Electrical, Computer and Energy Technologies (ICECET 2026), International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME 2026).

Conference Service: Volunteer, IEEE DySPAN 2026; Volunteer, International Chinese Statistical Association (ICSA) 2026.

Recognition: IEEE DySPAN Student Travel Grant (2025).

Leadership: Vice President (Administration), Bangladesh Graduate Student Association, George Mason University, Fairfax, VA, USA
Apr. 2024 – May 2025